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WILLIAM P. EVELAND, JR. and SHARON DUNWOODY

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WILLIAM P. EVELAND, JR.¹
SHARON DUNWOODY

User Control and Structural Isomorphism or Disorientation and Cognitive Load?

Learning From the Web Versus Print

User control theory predicts that providing freedom in learning increases learning compared to traditional instruction, implying that the Web is more effective for learning than print. Theorists have also argued that navigation through Web sites mimics the associative nature of human memory and information processing—structural isomorphism—suggesting Web superiority. However, studies indicate that hypermedia increases cognitive load and produces disorientation, implying that the Web would be less effective for learning than would print. An experiment comparing learning in print versus several Web site designs demonstrated that learning from print as measured by recognition is better than learning from linear and nonlinear Web designs but no different from a design including advisement. No significant differences across media conditions were found using cued recall as the measure of learning. Additional findings suggest that cognitive load inhibits learning, whereas Web expertise facilitates it. Curiously, a learning motivation tended to reduce learning.

During the past five years, access to and use of the World Wide Web has skyrocketed. Traditional mass media channels such as newspapers, television, and magazines are moving information wholesale to their Web sites. The ease and accessibility of the Web has encouraged everyone from affluent companies to iconoclastic individuals to create sites full of information relevant to the public.

Implicit in this rush to place information on the Web is the assumption that the Web is equal or superior to more traditional information channels for

the purpose of learning. But, moving information from traditional channels to the Web, especially when the learning of that information is important for health, education, or democracy, begs the question of whether the medium of mass communication makes a difference in the quality or quantity of learning. Given the practical issues involved, as well as the theoretical issues to be described below, the time has come to reconsider a basic but central question: Does learning differ when the information is presented via the Web versus a traditional print source, and are there variations in learning from the Web across different site designs? In this article, we present the results of an experiment designed to provide some evidence to help answer these questions.

Research Comparing Learning From Different Media

There is a long history of research comparing learning from different media, as researchers in mass communication (and other disciplines) have been concerned with the potential problems or excited by the potential benefits of using one particular medium or another for conveying information. Given changes in media over time, we will focus on research conducted during the past 30 years. This research has come from two separate paradigms that rarely reference one another: a survey paradigm and an experimental paradigm. Before examining the existing theory and research on learning from the Web versus print, we provide a brief overview of the broader research area of comparing learning from different media.

Survey Paradigm

Survey research comparing learning across different media has traditionally tested a random sample of individuals on their level of knowledge of a particular topic covered in the media, then assessed individual-level variation in use of different media. Correlations (or other statistics of association) are then computed to determine whether (a) exposure to a particular medium is related to knowledge of a topic covered in that medium; and (b) correlations are stronger between knowledge and use of one medium rather than an alternate medium.

Probably the most common context in the survey research paradigm is the comparison of learning from television news to that from newspapers (e.g., Chaffee & Schleuder, 1986; Clarke & Fredin, 1978; McLeod & McDonald, 1985; Robinson & Levy, 1986). This research has increased steadily during the past several years, sometimes even expanding to take into account learning from the radio or other news sources (e.g., Bennett, Flickinger, Baker, Rhine, & Bennett, 1996; Brians & Wattenberg, 1996; Chaffee, Zhao, & Leshner,

1994; Drew & Weaver, 1998; Martinelli & Chaffee, 1995; Price & Czilli, 1996; Robinson & Levy, 1996). Very recently, some researchers have begun to include measures of use of the Internet among their media measures (Johnson, Braima, & Sothirajah, 1999). Although it is still a contentious issue, we believe that the general conclusion from this body of research is that use of print is a better predictor of knowledge of media content than is use of television news or other news sources. This conclusion is supported by a recent meta-analysis of a number of survey studies (Halpern, 1997).

Experimental Paradigm

The second paradigm of research that examines learning from different media takes an experimental approach. This research, often comparing audio and audiovisual media to print, typically manipulates the medium of presentation by either transcribing existing audiovisual content (i.e., a television news story) to produce a print version (e.g., Gunter, Furnham, & Gietson, 1984; Gunter, Furnham, & Griffiths, 2000; Just & Crigler, 1989; Walma van der Molen & van der Voort, 2000), or less frequently, having a talking head narrate what originally was a print source on video (e.g., Furnham, Gunter, & Green, 1990). The bulk of this evidence indicates that when holding content constant, print is superior to audio-only or audiovisual presentations, regardless of the original source (e.g., Browne, 1978; Furnham et al., 1990; Furnham & Gunter, 1989; Furnham, Proctor, & Gunter, 1988; Gunter et al., 1984; Wilson, 1974). However, a few studies have found no differences between print and audiovisual presentations (e.g., Just & Crigler, 1989; Stauffer, Frost, & Rybolt, 1981) or that a print version was inferior to an audiovisual presentation (Just & Crigler, 1989).²

More recently, researchers in the experimental tradition have begun to examine learning from media other than television and print, often focusing on different types of computer-based technologies. Some have included learning from a computer screen in addition to television, print, and radio (DeFleur, Davenport, Cronin, & DeFleur, 1992), whereas others have even begun to examine learning from newspapers versus Web sites (Sundar, Narayan, Obregon, & Uppal, 1998; Tewksbury & Althaus, 1999). In addition to research by scholars in communication, research published in other fields has examined differences in learning from computers versus traditional print sources (e.g., Osborne & Holton, 1988; Psotka, Kerst, & Westerman, 1993; Reinking, 1988; Rice, 1994). One recent review of the research comparing hypermedia to print concluded that there is little evidence of a significant difference in learning across media (Dillon & Gabbard, 1998), but another (Chen & Rada, 1996) has come to a different conclusion.

Issues Confronting the Survey and Experimental Paradigms

The Survey Paradigm

Research from both the experimental approach and the survey approach to comparing learning across media has been problematic on both methodological and theoretical grounds. For the survey paradigm, the most difficult questions that must be answered are (a) How does one effectively measure the use of different media in comparable units (e.g., exposure time vs. quantity)? (b) When media naturally vary in their content, should learning be judged at the lowest common denominator (i.e., only the information contained in all media sources) to maintain comparability? and (c) What individual-level characteristics correlate (or interact) with media choice and exposure, and how can we partial out these influences to get at the true effects of the medium? Each of these questions will be addressed in turn.

COMPARABILITY OF UNITS

More than a decade ago, Chaffee and Schleuder (1986) argued that in making comparisons of learning from television news and newspapers, exposure did not mean the same thing for each medium. They argued that newspapers demand attention but that for television, attention was in effect optional. Thus, survey comparisons of learning from television and newspapers, they argued, should use measures of attention rather than exposure.

COMPARABILITY OF CONTENT

Comparisons of television news and newspapers are also difficult because of the large differences in content between the two. It is often noted that a standard evening news broadcast, if transcribed, could easily fit on the front page of *The New York Times* (e.g., Furnham et al., 1990). If this is so, then exposure to a full newspaper provides substantially more information than exposure to a half hour of television news. While this in part speaks to the issue of comparability of measures of exposure, it also begs the question, What content should be included in a knowledge test? To expect television viewers to know the details provided in a newspaper would be unfair if those details were not contained in the newscast. But, by the same token, to ask for recollection of only the main points as described in a television newscast disadvantages newspaper readers, who must contend with picking those details

out of a much larger base of information that might make recall more difficult even if the main points were initially processed and stored.

CHARACTERISTICS OF AUDIENCE

Finally, because education is more highly correlated with the use of print media than television news and also may help in the integration of new information into memory, it must be statistically controlled when comparing learning across media. This is true for a number of other variables that are correlated with both media use and knowledge acquisition. The problem is identifying these variables, determining that they are indeed spurious as opposed to having some theoretical import, and finally, accurately measuring and statistically controlling them.

The Experimental Paradigm

The problems with the experimental approach to comparing media for learning have been most clearly stated by educational technology researcher Richard Clark (1983, 1985; Clark & Salomon, 1986). The logic of Clark's criticism is simple. He argues that research comparing different media for learning tends to confound medium with content or, in his field of education, with instructional method. So, in conducting an experiment comparing one medium with another, if an effect is found for medium, it is actually an effect of some variable that is merely confounded with medium. Tightly controlled studies that vary only medium of presentation, he argues, will find no differences in learning across media. The exception is when differences occur due to a novelty effect of a new medium, in which case the effect will disappear over time. This again suggests that media do not have any independent effect on learning. In an oft-cited metaphor, Clark (1983) claimed,

The best current evidence is that media are mere delivery vehicles that deliver instruction but do not influence student achievement any more than the truck that delivers our groceries causes or changes our nutrition. Basically, the choice of vehicle might influence the cost or extent of distributing instruction, but only the content of the vehicle can influence achievement. (p. 445)

Such an argument probably raises the hackles of those who identify themselves as media effects researchers, and it made a substantial impact on those who study educational technology. A decade later, this argument was still so

central to the study of technology effects in the classroom that a special issue of *Educational Technology Research and Development* was devoted to a debate on the topic (e.g., Jonassen, Campbell, & Davidson, 1994; Kozma, 1994; Morrison, 1994; Reiser, 1994; Ross, 1994; Shrock, 1994; see also Ross & Morrison, 1993).

If one were to address Clark's (1983) criticism but still study media effects on learning, one would strive to eliminate all differences between conditions except the medium of presentation. Although this is of value for theoretical reasons and to those who are truly interested in the effect of a medium, it presents a problem for researchers with practical interests. To design a study in which only medium varied would produce a study that, although internally valid, would suffer from problems of external validity so badly that it would be almost useless for any practical applications. Simply put, content and other factors (sometimes called "formal features") (see Bucy, Lang, Potter, & Grabe, 1999 for a discussion in the Web context) often do, in the real world, covary with medium. More important, these covariations are interesting and important to many media researchers. To ignore this covariation in experiments would produce results that are, although technically clean, relatively meaningless.

Such are the results of much of the existing audiovisual-versus-audio-versus-print experiments. That is, when researchers transcribe a television newscast and present it to participants as printed text, does this really represent what most of us would be interested in—a newspaper or magazine story? The answer to this rhetorical question is "no" because a newspaper story is structured differently. It is longer and goes into more detail than a television news story. Similarly, television stories are typically accompanied by still or moving images. To remove these images to hold content constant makes the television condition unrealistic and thus the findings of the study less interesting.

In this article, we do not presume to provide a solution to these dilemmas. However, we discuss them here because they are important to consider in any study comparing learning from different media. Also, we believe that in the current context, comparing learning from the Web versus traditional print media, content can be held constant without rendering the comparison unrealistic, unlike most television-versus-print experiments. This is because the important theoretical differences between media in this study, as we discuss below, center not on the typical content of these two different media but on how that content is structured and how the freedom provided in the Web medium can either increase or inhibit learning.

Theory Relevant to Learning From the Web Versus Traditional Print

Despite the relatively recent arrival of the World Wide Web in the public consciousness, the technology behind the Web, labeled hypermedia, has received much theoretical attention (e.g., Ambrose, 1991; Bush, 1945; Heller, 1990; Morariu, 1988; T. H. Nelson, 1993) and empirical attention (Borgman, Walter, Rosenberg, & Gallagher, 1991; Gay, Trumbull, & Mazur, 1991; van den Berg & Watt, 1991). As a result, two (at least partially) competing hypotheses have been developed about learning from hypermedia. One hypothesis suggests that, for several reasons, the nature of hypermedia should allow greater learning than has before been possible, implying that learning from the Web should be greater than learning from traditional print. The alternate view is that the nature of hypermedia presents a hurdle for learners to overcome and thus could, at least in some cases, reduce learning compared to print presentations. These competing hypotheses are the focus of this study.

User Control

Educational psychologists have argued for some time now that providing control of the pace, order, and content of instruction to a learner can increase the amount of learning. This is true, they argue, for a number of reasons. First, individuals learn in idiosyncratic ways, and to force learning to take place in a single style will reduce the learning of some individuals to less than their true potential. Providing control and thus individualization of the learning process will allow these people to do better than they would in a cookie-cutter instructional setting (Kinzie, Sullivan, & Berdel, 1988; Young, 1996), improving overall learning.

Second, providing control over the learning process can increase the motivation of the learner (Kinzie, 1990; Steinberg, 1989). Especially in informal learning contexts such as the use of mass media, motivation to continue learning is central to learning itself. If one becomes bored with a topic, that individual is likely to stop reading or seeking information, precluding future learning. Thus, the control provided by hypermedia systems such as the Web to follow the paths of most interest and value and the paths that are at the appropriate level of expertise for the learner can increase motivation to learn, one's self-efficacy, and thus learning itself (Bandura, 1982; Schunk, 1991).

The theory of user (or learner) control has implications for the design of Web sites. Specifically, user control can be conceptualized as a continuum

ranging from complete control for the learner ("learner control") to complete control for the medium or site ("program control") (Milheim & Martin, 1991). The level of control afforded the learner is a direct function of the design of the Web site, with less structured sites providing more user control than sites with a clear structure, such as linear or hierarchically organized sites. In addition, some sites provide what has been labeled "instructional advisement," which is a limited form of control in that the site essentially makes suggestions about the most appropriate movements, but the user is free to override these suggestions at any time (Shin, Schallert, & Savenye, 1994). A guided tour with no real alternatives except for the choice of when to move to the next page would represent the minimum level of user control in a Web site. The appropriate level of user control to build into a Web site has become an empirical question and will be addressed in this study. If user control were not enough to argue for the superiority of learning from the Web versus traditional print, other arguments have been made that do not depend on the notion of user control.

Structural Isomorphism

A common view of the structure of human memory among psychologists is that bits of information are organized through their connections to each other. Research based on this perspective on human memory includes schema theory (Wicks, 1992), priming (Iyengar & Kinder, 1986), domains of memory (Tulving, 1985), connectionism (Smith, 1996), and levels of processing (Craik & Lockhart, 1972). Using this view of human memory, learning is in large part the process of creating and maintaining meaningful links among concepts (Jonassen, 1988; W. A. Nelson & Palumbo, 1992).

It has been argued that the process of using hypermedia—and thus the Web—is consistent with this common conception of the structure of human memory and the process of learning (e.g., Bieber, Vitali, Ashman, Balasubramanian, & Oinaas-Kullonen, 1997; Churcher, 1989; Jonassen & Wang, 1993; Kozma, 1987; Lucarella & Zanzi, 1993; Marchionini, 1988; Shirk, 1992). That is, both human memory and hypermedia are made up of individual nodes of information connected by links. Using hypermedia mimics the process of human learning because concepts discussed in one node or page of a Web site are linked to related nodes discussing related concepts.

Furthermore, the argument made by many hypermedia advocates is that because hypermedia can be designed to emulate the appropriate links among concepts in a particular knowledge domain, learners can develop better knowledge structures from hypermedia than from traditional, linear print

texts. However, some theorists have taken issue with the claim of isomorphism between hypermedia systems and human memory (W. A. Nelson & Palumbo, 1992; Tergan, 1997), in part because the process of linking in human memory is much stronger than the generic, often ambiguous links found on the Web.

Disorientation and Cognitive Load

Despite claims to the contrary, some theorists and researchers have noted that hypermedia systems such as the Web may actually reduce learning by increasing cognitive load and producing disorientation among users, particularly those with little experience in the medium or content domain. In the context of learning from media, *cognitive load* may be defined as the amount of mental effort required to locate specific information and understand how this information is oriented within a larger information source (i.e., a section of a magazine article or a page on a Web site). Generally speaking, hypermedia use imposes more cognitive load than the use of printed text because hypermedia use requires "a certain overhead of metalevel decision making, an overhead that is absent when the author has already made many of these choices for you" (Conklin, 1987, p. 40). Different hypermedia structures, particularly those with increased user control, could exacerbate cognitive load (Oliver, 1996).

Navigation through Web sites with high degrees of user control, making use of nonlinear document structures, has been likened to navigation through unfamiliar physical space. Indeed, much of the research on hypermedia systems and the Web has described user feelings of being "lost" in hyperspace (Conklin, 1987) or, more generally, "disorientation" (e.g., Calvi, 1997; Darken & Sibert, 1996; McDonald & Stevenson, 1996). These feelings may be caused by both inexperience with the medium or content and site design. In any case, disorientation will increase the cognitive load required to successfully navigate the site.

It is assumed by many psychologists that the capacity of humans to process information is finite and is often exceeded by the information available in the environment (e.g., Lang, 2000).³ To deal with this limitation in cognitive capacity, users may not be able to devote an optimum amount of cognitive effort to all tasks. Therefore, if a Web site's design requires substantial cognitive effort simply to navigate through it (cognitive load), less effort will be available to invest in actual learning processes such as integration or elaboration (Eveland & Dunwoody, 2000b). Because these processes are very important components of the act of learning, site designs that provide too

much user control—or that eschew cues of a traditional linear structure—can reduce learning. A number of studies have demonstrated that tasks high in cognitive load can inhibit learning (Mayer, Moreno, Boire, & Vagge, 1999), at least among those with limited expertise (Tuovinen & Sweller, 1999). But, even setting immediate learning aside, if a site is too complex and produces disorientation, it might discourage individuals from ever returning or from seeking information elsewhere in the future. This may be an even more dangerous effect of poor site design and high cognitive load than reduced immediate learning.

In summary, from the perspective of disorientation and cognitive load, at least some Web site designs may reduce learning and motivation compared to a print version. These hypotheses are directly opposed to the predictions that stem from the theory of user control.

Hypotheses

Based on the review of the literature above, six hypotheses were tested. Two of these are competing hypotheses derived from the literature on learning from hypermedia and the World Wide Web. The following hypotheses stem from the perspective of user control and theories of structural isomorphism between hypermedia navigation and human information processing:

Hypothesis 1a: Learning will be greater for the Web than for print, holding content constant.

However, theories of disorientation and cognitive load offer a competing hypothesis:

Hypothesis 1b: Learning will be greater for print than for the Web, holding content constant.

The third hypothesis is derived from the literature on both user control and disorientation and cognitive load:

Hypothesis 2: Learning from the Web will be greatest when both user control and instructional advisement are provided, compared to either component alone.

In addition to these three hypotheses, several additional hypotheses can be offered. Based on the user control literature and research on learning from media, we can expect that a learning motivation will be positively related to

actual learning (McLeod & McDonald, 1985; Neuman, 1976), although recent research indicates that this relationship is mediated by media information processing variables such that motivations increase learning indirectly by encouraging information processing (see Eveland, 1997, 2000). Therefore, we predict the following:

Hypothesis 3: Learning in all conditions will be greater among those with stronger learning motivations.

Early hypermedia scholars argued that “expert users who are specialists in the task domain will welcome the great power and control [of hypermedia], but novices to the system and task domain will likely benefit from limited menus and less [user] control” (Marchionini & Schneiderman, 1988, p. 78). In addition, several recent knowledge gap studies (Eveland & Scheufele, 2000; Grabe, Lang, Zhou, & Bolls, 2000) focus on various information processing skills as explanations for differences in learning from media. If expertise using a medium like the Web is an information processing skill—and we would argue that it is—then Web expertise should play an important role in learning, at least from the Web. Based on these claims, we could expect that Web expertise would moderate the effect of medium such that Web experts might reveal a weaker difference across conditions than would Web novices because the Web experts’ skills can compensate for potentially increased task complexity. There should also be a main effect of Web expertise in the three Web conditions. Therefore, we predicted the following:

Hypothesis 4: There will be an interaction between Web expertise and medium such that there will be stronger differences across conditions for Web novices than for Web experts.

It is likely that in addition to the cognitive load demands created by the formal features of a medium and its structure, individuals will vary in their perceptions of the complexity of the task to which they are assigned. This implies that, in addition to task-imposed cognitive load, individuals will subjectively experience different levels of cognitive load. Thus, we would expect that cognitive load as experienced and perceived by the individual would reduce learning because it would reduce the amount of cognitive effort devoted to learning processes. Thus, we predict the following:

Hypothesis 5: Cognitive load will be negatively related to learning in all media conditions.

Method

Participants

Participants were recruited from a number of courses across two departments at a large, Midwestern university in the spring of 1998. Approximately one half were freshman, 22% were sophomores, 13% were juniors, 11% were seniors, and 4% were graduate students. Nearly two thirds of participants were women, and their average age was 20.44 years ($SD = 2.72$). This young, relatively educated group of participants was important to this study because they had experience using the Web. They reported using the Web an average of 12.84 ($SD = 9.96$) days during the past 30 days, and only 4% claimed they had not used the Web in the past month.

Procedures

Initially, the study took a 4 (medium) $\times 2$ (motivation) $\times 2$ (elaboration) $+ 1$ (control) between-subjects design, but some cells have been collapsed for this report.⁴ Thus, for the purposes of this report, there were five cells: control ($n = 28$), print ($n = 50$), linear Web ($n = 48$), nonlinear Web ($n = 46$), and advisement Web ($n = 47$).

After signing up for the study, participants were directed to report to a research room on campus. Within each time slot, individuals were randomly assigned to one of the conditions. Those assigned to the control and print conditions were directed to a classroom with large desks for the remainder of the study. Those assigned to one of the three Web conditions were directed to a classroom adjacent to that for the print and control conditions. This classroom was a computer laboratory equipped with 16 Macintosh computers with grayscale, page layout style monitors.

The relevant content for this study concerned the flu virus, a topic chosen because of its relevance during the time the study was conducted (flu season). It initially appeared on the Web science magazine *The Why Files* (see <http://whyfiles.org/049flu/index.html> for the original story) and was a combination of text, photographs, and other graphics. The story consisted of an index page (describing the content of the story and providing links to different sections), six pages of content, a bibliography, and a glossary of terms. For purposes of the experiment, the story was modified to create three Web versions: linear, advisement, and nonlinear.⁵ These versions varied by the availability of navigation options between pages of the text, but the content remained the same.

In the linear Web version, aside from links to the bibliography and glossary where appropriate, the only links in the text were at the end of each page, and they directed the user to the next contiguous page. These links took the form of verbal teasers, which encouraged users to follow the story in linear order. However, at the bottom of each page were numbered links that allowed the user access to all other pages in the story, if desired,⁶ and “next,” “back,” and “story map” (index) links to ease navigation. Thus, although linearity was encouraged in this version of the Web site, it was not absolutely required like it would have been in a guided tour design. Thus, this site represented a very weak form of user control beyond a guided tour.

In the nonlinear Web version, in-text links were added where appropriate to the linear version to allow users to move more freely nonlinearly through the story. So, instead of waiting to finish a page of reading, users could jump from one page to another by following a highlighted link in the text, providing a high degree of user control. In addition, the teaser links were retained. However, the numbered links and navigation buttons at the bottom of the page were removed. This compelled the user to navigate via potentially nonlinear in-text links because there were no clear linear options available.

In the advisement Web version, all the navigation options available in the nonlinear Web version were maintained. In addition, the navigation options at the bottom of the page that were part of the linear Web version—the teaser, plus numbered page links and “next,” “back,” and “story map” buttons—were available, providing advisement on navigation through the story.

In the print condition, participants received a grayscale paper copy of the story from the Web site. All links and navigation buttons were removed, including indications that certain words were in the glossary. The text and graphics from the Web site were merged into a single, continuous document analogous to a print magazine in the same order provided in the linear Web condition. The index page was redesigned as a table of contents with page numbers replacing the direct links. These changes allowed us to hold the content of the story constant relative to the Web condition. The only difference was in the means of navigation, particularly given the use of a grayscale page view monitor for the Web conditions.

In the control condition, participants were provided with a black and white print article that was unrelated to the flu virus content given to those in all other conditions. In this manner, control participants spent the same amount of time in the laboratory as others but were not exposed to the content on which the posttest quiz was based.

The experiment was timed so that participants spent equal amounts of time (15 minutes) on the stimuli both across and within conditions. At the end of the study, all participants received the same posttest questionnaire.

Measurement

A posttest questionnaire measured a number of variables used in this study (see appendix). Covariates included learning motivations, Web expertise, and cognitive load. Learning motivations ($M = 9.68$, $SD = 2.18$) were measured with three 5-point Likert-type items ($\alpha = .68$).⁷ Web expertise was an additive index of two items ($r = .71$), one for the number of days in the past 30 in which the participant had used the Web ($M = 12.84$, $SD = 9.96$), and the other a single item self-report of Web expertise using a 10-point scale ($M = 5.16$, $SD = 2.23$). Because these items were measured on different scales, they were standardized before scale construction. Cognitive load ($M = 8.61$, $SD = 2.90$) was measured with four 5-point Likert-type items (two positively worded, two negatively worded) on which respondents indicated difficulty recognizing the organization and structure of the information ($\alpha = .79$).

Lang (2000) has argued that "different measures of memory can be used to index different degrees of memory, which, in [the limited capacity model of mediated message processing] is interpreted as the thoroughness of various subprocesses" (p. 56). She maintains that recognition is a measure of encoding of information, cued recall is a measure of thoroughness of storage, and free recall is a measure of retrieval processes. Similarly, Shoemaker, Schooler, and Danielson (1989) have argued for the importance of distinguishing between recognition and recall measures for comparing effects across media.

In this study, participants were given a quiz on the content of the flu article that included a number of items in matching, multiple choice, fill in the blank, true/false, check all that apply, and short answer formats. Given possible differences across types of memory, we constructed two knowledge measures using these indicators: Recognition was a 15-item additive index ($\alpha = .61$) of matching, multiple choice, true/false, and check all that apply items ($M = 10.31$, $SD = 2.59$); cued recall was a 9-item additive index ($\alpha = .70$) of fill in the blank and short answer items ($M = 5.85$, $SD = 2.50$).⁸

Results

The results are organized according to the dependent measure: either recognition or cued recall. Identical analyses were conducted for each dependent variable.

Table 1
General Linear Model Analysis of Predictors of Recognition and Cued Recall

	Type III SS	df	MS	F	B	p	Partial η^2
Recognition							
Corrected model	151.78	6	25.30	5.12	—	.00	.143
Intercept	1209.35	1	1209.35	244.80	13.42	.00	.571
Learning motivation	31.94	1	31.94	6.47	-0.19	.01	.034
Web expertise	23.97	1	23.97	4.85	0.19	.03	.026
Cognitive load	43.48	1	43.48	8.80	-0.16	.00	.046
Media manipulation ^a	52.51	3	17.50	3.54	—	.02	.055
Error	909.00	184	4.94				
Total	22892.00	191					
Corrected total	1060.78	190					
Cued Recall							
Corrected model	126.79	6	21.13	5.27	—	.00	.147
Intercept	542.77	1	542.77	135.31	8.70	.00	.424
Learning motivation	15.65	1	15.65	3.90	-0.14	.05	.021
Web expertise	22.45	1	22.45	5.60	0.19	.02	.030
Cognitive load	60.92	1	60.92	15.19	-0.19	.00	.076
Media manipulation ^a	27.83	3	9.28	2.31	—	.08	.036
Error	738.07	184	4.01				
Total	8632.00	191					
Corrected total	864.86	190					

Note. Recognition $R^2 = .143$ (adjusted $R^2 = .115$). Recall $R^2 = .147$ (adjusted $R^2 = .119$).

a. This includes print, linear Web, nonlinear Web, and advisement Web conditions, but not the no exposure control condition.

Recognition

A first step in the analysis was to demonstrate that learning took place in the experimental conditions compared to the control condition using the recognition measure. To do so, an ANOVA was conducted with the five-level media manipulation variable. Results indicated significant differences across conditions, $F(4, 214) = 12.44$, $\eta^2 = .189$, $p < .001$, and Dunnett's test revealed that all experimental conditions were significantly different from the control condition (all $ps < .001$). All further analyses were conducted excluding participants in the control condition (recognition $M_{\text{control}} = 7.68$), leaving the one print and three Web conditions.

To test the hypotheses of the study, an analysis using the General Linear Model (GLM) was conducted with learning motivations, Web expertise, and cognitive load as covariates and the media manipulation as a between-subjects experimental factor. The results of this analysis are reported in Table 1, and means are presented in Figure 1.

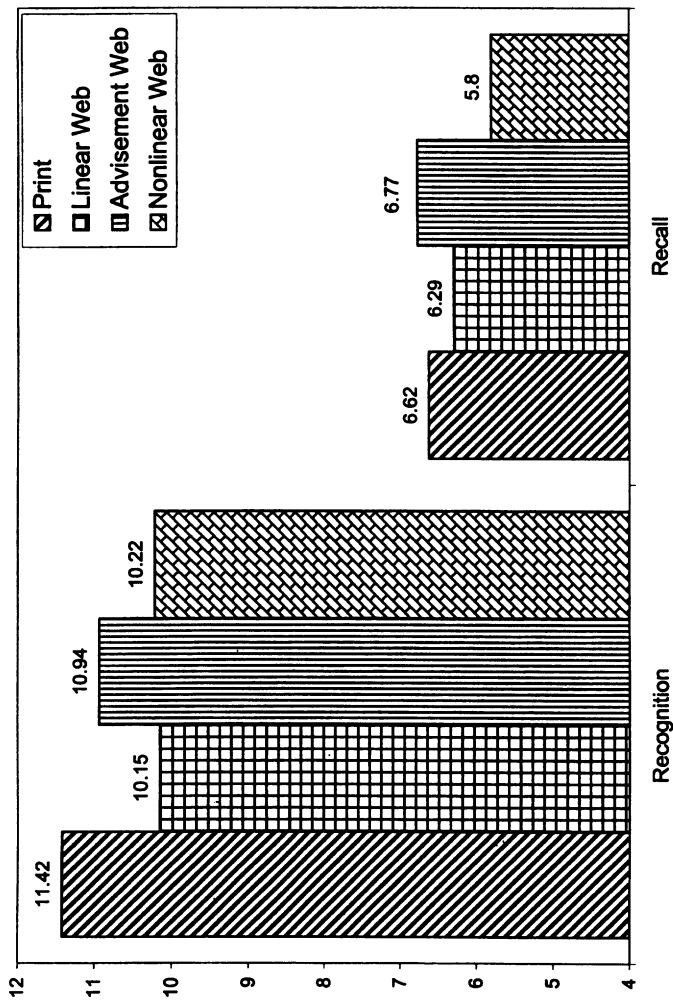


Figure 1. Recognition and Cued Recall Means, by Medium

All variables accounted for significant variance in recognition scores. The media manipulation produced a significant effect, accounting for more variance than any of the covariates individually. To test Hypotheses 1a, 1b, and 2, post hoc analyses were conducted using Tukey's honestly significant differences (HSD) test. These analyses were conducted without covariates and indicated that only one significant difference emerged. Recognition from traditional print was significantly greater than from the linear Web version ($p < .05$). In addition, the difference between traditional print and the nonlinear Web condition approached conventional levels of statistical significance ($p = .054$). These findings and the plot of means in Figure 1 provide limited support for Hypothesis 1b—that learning would be greater for print than the Web conditions—but no support for Hypotheses 1a or 2.

Moving to the impact of covariates in the GLM analysis, learning motivations were negatively related to recognition, indicating that those who most strongly agreed that they were reading to learn actually learned less than those who were less likely to claim learning goals. This significant finding is inconsistent with Hypothesis 3. This finding was not qualified by an interaction with the medium manipulation, $F(3, 184) = 1.61$, $\eta^2 = .026$, $p = .19$.

Hypothesis 4 predicted that the impact of the medium manipulation would be greater among Web novices than Web experts. However, the interaction between medium and Web expertise was not significant, $F(3, 184) = .42$, $\eta^2 = .007$, $p = .74$. Despite this, the main effect of Web expertise was significant, demonstrating that Web experts learned more in all media conditions (including the print condition) than did those who were novices.

Hypothesis 5 was supported by the finding that those who reported greater cognitive load scored lower on the recognition test. This finding was not qualified by an interaction with the media manipulation, $F(3, 184) = .54$, $\eta^2 = .009$, $p = .66$.

Cued Recall

As for the recognition measure, the first step was to demonstrate that learning took place in the experimental conditions compared to the control condition using the cued recall measure. To do so, an ANOVA was conducted with the five-level media manipulation variable. Results indicated significant differences across conditions, $F(4, 214) = 26.06$, $\eta^2 = .319$, $p < .001$, and Dunnett's test revealed that all experimental conditions were significantly different from the control condition (all $ps < .001$). All further analyses using cued recall were conducted excluding participants in the control condition (cued recall $M_{\text{control}} = 2.29$), leaving the one print and three Web conditions.

As for the recognition measure, an analysis using GLM was conducted with learning motivations, Web expertise, and cognitive load as covariates and the media manipulation as a between-subjects experimental factor. The results of this analysis are reported in the bottom half of Table 1, and means are presented in Figure 1.

The media manipulation approached but did not reach conventional levels of significance. Furthermore, tests comparing the means using Tukey's HSD test indicated that none of the differences in means across the media conditions approached statistical significance ($p > .10$). Thus, Hypotheses 1a, 1b, and 2 were all rejected. There were no significant cued recall differences across media conditions.

Moving to the impact of covariates in the GLM analysis, learning motivations were negatively related to recognition, indicating that those who most strongly agreed that they were reading to learn actually learned less than those who were less likely to claim learning goals. This significant finding is inconsistent with Hypothesis 3. This finding was not qualified by an interaction with the medium manipulation, $F(3, 184) = .79$, $\eta^2 = .006$, $p = .79$.

Hypothesis 4 predicted that the impact of the medium manipulation would be greater among Web novices than Web experts. However, the interaction between medium and Web expertise was not significant, $F(3, 184) = .79$, $\eta^2 = .013$, $p = .50$. Despite this, the main effect of Web expertise was significant, demonstrating that Web experts learned more in all media conditions (including the print condition) than did those who were novices.

Hypothesis 5 was supported by the finding that those who reported greater cognitive load scored lower on the recognition test. This finding was not qualified by a significant interaction with the media manipulation, although the interaction did approach significance, $F(3, 184) = 2.56$, $\eta^2 = .041$, $p = .06$.

Discussion

The findings of the study described in this article contribute to the experimental literature comparing learning from hypermedia or the Web to learning from print, as well as to the larger literature using both survey and experimental methods to compare learning across numerous media. As an experiment, the study presented in this article permits inferences of causality by controlling extraneous factors, including variations in content across media. There is little concern here, as there would be in survey research, that more knowledgeable individuals or individuals more successful at learning generally were selectively using one medium over another. Of course, unlike survey research, this study focused on a homogeneous population (college

students) using media in a controlled environment and thus is less ecologically valid.

Despite the high degree of experimental control in these studies, Clark (1983, 1985) might still argue that our finding of greater recognition from print than from two of three versions of a Web site is attributable to something other than medium of presentation. That is, Clark would likely claim that our study was guilty of the media comparison confound. We acknowledge this critique, but at the same time we must point out that it is not the medium per se that is most interesting in this study but the organizational structure of the content and various reading patterns that are facilitated by the medium that are most interesting and also at the core of our findings. The theoretical approaches from which we derived our hypotheses are clear on this point as well.

Our findings provide limited evidence consistent with the claim that hypermedia presentations may reduce learning compared to traditional print because of factors such as cognitive load and disorientation. In addition to the significant difference in recognition and marginally significant difference in cued recall across media conditions, perceived cognitive load accounted for a significant increment in variance in both learning measures. Similarly, Web experts learned more than Web novices regardless of medium, although the expected interaction between expertise and medium did not materialize. Somewhat surprisingly, learning motivations were associated with less learning, not more, as predicted.

Medium Differences and Cognitive Load

Our initial expectation when conducting this study was that the measure of cognitive load could be used as a manipulation check for the medium manipulation. However, there were no significant differences across the four media conditions in self-reported cognitive load. Similarly, Web expertise was not significantly related to perceptions of cognitive load. These findings could mean one of two things. First, they could suggest that our measure of cognitive load did not actually tap cognitive load. Second, they could mean that our medium manipulation was not sufficient to overcome individual differences in cognitive load perceptions.

Although we used theory to guide us, our measure of cognitive load has not been tested or validated before. Other researchers have used either secondary response time (Harms, 1991) or physiological responses (Wickens, 1992) to measure cognitive load, and some simply assume it is produced by

variations in the complexity of a given task (e.g., Mayer et al., 1999). Although there has been some effort in engineering to measure similar constructs with survey measures (see Sheridan, 1992), we did not find any of these measures to be conducive to our context. The significant negative relationship between self-reported cognitive load and learning on both recognition and cued recall measures in our study suggests that our survey measure does have some validity. However, we think it best for now to interpret this measure as self-perceived cognitive load, which is not exactly the same as actual cognitive load.

If the lack of differences in self-perceived cognitive load across media conditions is due to a relatively weak experimental manipulation of this factor, then we must develop alternate explanations for the differences in learning across conditions. Given that we limited participants to 15 minutes for exposure—intentionally too short a time for most to complete the reading—the facilitation of reading speed or comprehension by medium might help explain our findings. Unfortunately, we were unable to directly measure how much was being read across conditions, so a firm acceptance or rejection of this explanation must be left to further research. However, we did include a self-report measure of “selective scanning” derived from survey research on media information processing strategies (Kosicki & McLeod, 1990).⁹ We found that there was a significant difference in the extent of selective scanning across conditions, $F(3, 187) = 5.21$, $\eta^2 = .077$, $p < .01$, with the least scanning in the print condition, and significantly more (using Tukey’s HSD, $p < .05$) scanning in the advisement Web and nonlinear Web conditions. This suggests that actual reading—as opposed to simply scanning or browsing—may be discouraged as Web designs offer more nonlinear options. However, despite the fact that several studies have indicated that selective scanning is negatively related to measures of knowledge and learning (e.g., Eveland & Dunwoody, 2000a; Eveland & McLeod, 1995), the relationships between selective scanning and both recognition ($r = -.08$, $p = .28$) and recall ($r = -.12$, $p = .09$) in this study were in the correct direction but nonsignificant.

More generally, the validation of a survey measure of cognitive load beyond what we have done here should be a focus of future research efforts at understanding the psychological processes underlying learning from the Web versus print. An alternative to the self-report approach would be to make use of think-aloud interviews to assess the relative quantity of orientation thoughts to thoughts designed to integrate the content with background knowledge (Eveland & Dunwoody, 2000b) and compare these across different media or Web site designs.

Web Expertise

We expected, consistent with the logic utilized in recent knowledge gap research (Eveland & Scheufele, 2000; Grabe et al., 2000) and theories of hypermedia use (Marchionini & Schneiderman, 1988), that Web expertise as an indicator of information-processing skill would moderate learning across different media conditions. Specifically, we expected that there would be reduced differences in learning across media among Web experts relative to novices because expertise would dull the impact of complex Web designs on learning. Although there was a main effect of Web expertise on learning, the interaction between Web expertise and medium was not significant. Thus, Web expertise contributed to increased learning regardless of the medium of communication.

It may be that Web expertise is highly correlated with print expertise. That is, development of expertise in the Web medium, which is often merely an electronic form of traditional print, may lead to more effective use of traditional print as well. This is supported by recent findings that "the Internet is emerging as a supplement to—not a substitute for—other traditional news sources" (Pew Research Center, 1998).

Learning Motivations

Some research has demonstrated that learning motivations are correlated with information learned from media (McLeod & McDonald, 1985; Neuman, 1976). Elsewhere, we have argued that this relationship is indirect and mediated by information-processing activities (Eveland, 2000). Some recent experimental studies have not found significant differences between individuals told to expect a test versus those not so instructed (e.g., Gunter et al., 2000; Walma van der Molen & van der Voort, 1997). However, we were surprised to find in this study that self-reported learning motivations were negatively related to learning; that is, those who claimed to try to learn actually learned less than those who said they did not attempt to learn.

One explanation for this finding is that individuals do not always engage in effective strategies for learning despite their best efforts. Psychologists remind us that "the operations carried out on the material, not the intention to learn, as such, determine retention" (Craik & Tulving, 1975, p. 269). Anderson (1980) similarly notes that "whether or not one intends to learn or not really does not matter. What matters is how one processes the material during its presentation" (p. 197). In fact, in some studies in which the processing in incidental and intentional learning conditions is not strictly controlled,

participants may learn less in the intentional condition because “some of them were busy employing poor memorization strategies” whereas “the incidental subjects simply elaborated on the sentences and were not hampered by poor theories of what makes for good memory” (Anderson, 1980, p. 198). Thus, it is possible that the effort implied by our participants’ claim of intention to learn may have been focused on inappropriate or inefficient strategies for learning. Thus, the greater effort one expended in learning (using a poor learning strategy), the poorer the learning outcome.

Recognition Versus Cued Recall

We found few differences in influence for our independent variables between the recognition and cued recall measures of learning. For both measures, the learning motivation, Web expertise, and cognitive load variables produced similar patterns and magnitudes of relationship. The only difference was that the medium manipulation was statistically significant for recognition but only approached significance for the cued recall measure. The change to nonsignificance for cued recall reflects proportionately weaker learning in the print conditions compared to other conditions (particularly the linear and advisement Web conditions), or alternatively a relative increase in learning for the Web conditions.

Lang’s (2000) limited capacity model of mediated message processing provides a framework in which to interpret these findings. This model suggests that encoding of the information in the print condition was facilitated compared to the linear and nonlinear Web conditions but that this advantage in encoding was not carried over in terms of storage. Because the encoding stage involves exposure and attention, it may be that the print condition did facilitate a focus on the content of the story and perhaps greater overall exposure because of increased reading speed. However, the process of storage did not privilege the print condition. This indicates that the presence of links between pages and the requirement of thinking about navigation through the information space may have allowed participants in the Web conditions to catch up; alternatively, a lack of this sort of processing might have led the print condition to lose its advantage for learning.¹⁰

Although the distinction between recognition and recall may have some importance, we must acknowledge that our conclusions may have been different had we used yet another, more complex form of knowledge measure. Generally speaking, our tests tapped content knowledge, whether through recognition or cued recall. But, a measure of structural knowledge (Goldsmith, Johnson, & Acton, 1991; Joiner, 1998; Jonassen, Beissner, & Yacci, 1993;

Shavelson, 1974) may have produced quite different results. This is because the unique opportunities for communicating the structure as well as the content of knowledge using nonlinear hypermedia formats could encourage more sophisticated—that is, more differentiated and tightly integrated—knowledge structures more so than simply adding to content knowledge. Future research should attempt to test this hypothesis by measuring not only recognition and recall but also structural knowledge as conveyed by the organization of media content.

Conclusions

Media practitioners interested in communicating factual information to the public may consider these findings a bad omen for the swift movement of the role of information dissemination to the World Wide Web. However, given the relatively small effect sizes we found, other factors beyond learning potential should also be considered in the choice of medium for information dissemination. Harkening back to Clark's (1983) metaphor of the delivery truck can be useful here. He argued that the delivery truck can influence the cost or breadth of food distribution but that it does not directly affect nutrition. Although we take issue with this claim, the importance of choice of medium for speed, breadth, and cost of delivery are in fact important pragmatic issues to consider. The cost of producing a Web magazine compared with a comparable print magazine is clearly in favor of the Web magazine (Eveland & Dunwoody, 1998a). This is also true for the speed of production, as posting content to the Web is essentially immediate whereas the printing and physical distribution of a magazine is more time consuming. Thus, despite the potential for a slight reduction in learning from a linear or nonlinear Web site versus a print version of a magazine, other factors suggest that using the Web to disseminate information may not be a bad decision after all, as long as the current restriction in breadth of access to the World Wide Web is taken into consideration.

Appendix *Measurement*

Surveillance Gratifications (5-point Likert-type items)

I read the story with the goal of helping me decide my position on issues of public health.

I read the story with the goal of getting information I can use in my everyday life.

I read the story with the goal of learning as much as possible.

Web Expertise

About how many days out of the past 30 days have you used the World Wide Web?
If 1 were to represent a Web novice (beginner) and 10 were to represent a Web expert,
what number would you say represents your experience level in using the World
Wide Web?

Cognitive Load (5-point Likert-type items)

I had difficulty understanding how the information was structured into a coherent
story.

Sometimes I felt "lost" when reading the story.

The main points of the story were clear and coherent. (reversed)

It was clear how all the information fit into the story as a whole. (reversed)

Notes

1. The research reported in this article was supported by a cooperative agreement between the National Science Foundation and the University of Wisconsin–Madison (Cooperative Agreement No. RED-9452971). At the University of Wisconsin–Madison, the National Institute for Science Education is housed in the Wisconsin Center for Education Research and is a collaborative effort of the College of Agricultural and Life Sciences, the School of Education, the College of Engineering, and the College of Letters and Science. The collaborative effort is also joined by the National Center for Improving Science Education, Washington, D.C. Opinions, findings, and conclusions in this article are those of the authors and do not necessarily reflect the views of the supporting agencies.

2. Recent studies among both children and adults have found that the superiority of print may be replaced by the superiority of television using video presentations of news that are designed for children and employ reinforcing visual and verbal messages (e.g., Gunter, Furnham, & Griffiths, 2000; Walma van der Molen & van der Voort, 1997). However, among adults viewing traditional television news (which often has noncorresponding visual and verbal messages), these researchers have still found the superiority of print media (Walma van der Molen & van der Voort, 2000).

3. Fiske and Taylor (1991) distinguish between the "cognitive miser" model and the "motivated tactician" model. Both acknowledge the limited information-processing capacity of individuals, but the motivated tactician model also considers motivation to be important, particularly in the context of decisions about how information processing capacity will be allocated to various tasks. We subscribe to the motivated tactician perspective.

4. We focus in this article on the medium manipulation and ignore both the elaboration and motivation manipulations. The reason is that manipulation checks on the elaboration and motivation manipulations indicated no significant differences across conditions, no effects on the dependent measures, and no interactions between medium and the elaboration and motivation factors. This leads us to believe that we did not successfully manipulate these two variables in this study.

5. These general design categories, as well as others such as hierarchical structures, have been suggested by others (e.g., Korthauer & Koubek, 1994; McDonald & Stevenson, 1996, 1998; Shapiro, 1999; Shin, Schallert, & Savenye, 1994; Yang & Moore, 1995-1996) and reflect a broad cross-section of Web sites designed for information dissemination. Other characteristics of our experimental Web site designs—the lack of

video and audio content and the presence of hypertext links using text and graphics—are typical of the most popular sites on the Web (Bucy, Lang, Potter, & Grabe, 1999).

6. Although the page number links provided access to pages in a nonlinear fashion, they were not very conducive to this function. The links provided no information about the content of the given pages, only their order within the structure, making them in some sense blind links. In addition, research on the use of these links among natural users of the site outside of the experimental context indicated these links are used infrequently (Eveland & Dunwoody, 1998b).

7. Much like other studies (e.g., Walma van der Molen & van der Voort, 1997), learning motivation was manipulated in the experiment by a single sentence in the instructions. For the intentional learning condition, the instructions included the following sentence: "As you read this story, we'd like you to be prepared to answer questions later about your recall and understanding of the story content." For the incidental learning condition, the instructions included the following sentence: "As you read this story, we'd like you to be prepared to answer questions later about how interesting and entertaining you think the story is." However, the manipulation check (the learning motivation scale) revealed no significant differences in learning motivation across conditions, and a test of the effects of this manipulation revealed no significant effects. Therefore, we ignore the experimental manipulation and analyze the manipulation check self-report instead.

8. These means include participants in the control condition, who were not exposed to the relevant information.

9. Selective scanning ($r = .47$) was measured using two items measured on 5-point Likert-type scales: "I only read sections that looked important" and "I skimmed through the story."

10. Another factor in producing these differences could be derived from the encoding specificity principle (Tulving & Thomson, 1973). This principle suggests that when the same contextual cues are available at both encoding and retrieval, memory should be improved. Because all learning tests were conducted on paper, there were stronger cues for those initially exposed to the print condition than those in the Web condition, thus potentially benefiting memory in the print condition. Because the same pattern was found for both recall and recognition (but was nonsignificant for recall), this explanation could also generally account for our findings. We would like to thank an anonymous reviewer for suggesting this explanation.

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